

A Computer-Assisted Proof that $k(18) = 34$

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Abstract

Let $k(18)$ be the smallest number of distinct positive integer denominators, all at least 18, whose reciprocals sum to one. We establish $k(18) = 34$ by an exact 34-term identity and a computer-assisted exclusion of every shorter length. A harmonic bound excludes lengths at most 30. Complete rational searches exclude lengths 31 and 32; a complete 33-term search uses necessary prime-power coverage bounds with upward integer rounding. Its full replay is supplemented by separately implemented terminal verification. We give the coverage argument, arithmetic safeguards, computation counts, and the precise verification boundary. The conclusion concerns this finite minimization problem only; it neither settles the associated asymptotic question nor establishes historical priority.

1 The finite problem and its context

For $t \geq 1$, put

$$S_t = \left\{ (n_1, \dots, n_t) \in \mathbb{Z}_{>0}^t : 18 \leq n_1 < \dots < n_t, \quad \sum_{i=1}^t \frac{1}{n_i} = 1 \right\}.$$

Define $k(18) = \min\{t \geq 1 : S_t \neq \emptyset\}$; the identity below establishes that this minimum exists. The first denominator is allowed to exceed 18. Repetition, infinite expansions, and nonintegral denominators are outside this definition. Sorting preserves every admissible set of denominators.

The dated online report [1] reports the interval $33 \leq k(18) \leq 35$ on 11 August 2026, describes exact exclusions at lengths 31 and 32, and displays a 35-term expansion. Its linked verification bundle was unavailable during our citation audit, so the reported exclusions are cited as the report's assertions, not as independently verified external computations. This reported interval motivated the present calculation; our lower bound instead rests on the retained project computations below. The report also discusses whether $k(N) - (e - 1)N$ tends to infinity. It attributes that question to Erdős and lists a book by Erdős and Graham among its sources. We describe the report's attribution; the original book page was not directly inspected during the project. Our conclusion does not answer the asymptotic question. A limited literature search does not establish priority or certify a current open-status claim.

Theorem 1. *The minimum number of distinct positive integer denominators at least 18 whose reciprocals sum to 1 is 34. Equivalently,*

$$k(18) = 34.$$

The lower bound is computer-assisted. Its mathematical component proves finite coverage and the safety of branch deletions. Its computational component is the completed traversal of the resulting finite trees. The full 33-term replay uses the reviewed search implementation; it is not a second independent implementation of that whole search.

2 An exact upper certificate

Let W be the following ordered list, displayed over several lines:

18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34,
 35, 36, 38, 39, 40, 43, 44, 45, 46, 48, 50, 52, 54, 63, 275,
 14143068, 10457416992.

There are 34 entries, strictly increasing and at least 18. Exact integer calculation gives

$$L = \text{lcm}\{d : d \in W\} = 6209352774424800, \quad \sum_{d \in W} L/d = 6209352774424800 = L.$$

Every d divides L , so division by L proves $\sum_{d \in W} 1/d = 1$ and hence $k(18) \leq 34$. This certificate can be verified by the following complete Python code, independently of its discovery:

```
from math import gcd
w = [18,19,20,21,22,23,24,25,26,27,28,29,30,31,32,33,34,
     35,36,38,39,40,43,44,45,46,48,50,52,54,63,275,
     14143068,10457416992]
assert len(w) == 34 and all(type(d) is int for d in w)
assert w[0] >= 18 and all(x < y for x,y in zip(w,w[1:]))
L = 1
for d in w:
    L = L // gcd(L,d) * d
assert L == 6209352774424800
assert all(L % d == 0 for d in w)
assert sum(L // d for d in w) == L
```

3 A complete finite recurrence

Lemma 2. *No member of S_t exists for $1 \leq t \leq 30$.*

Proof. Strict increase gives $n_i \geq 17 + i$. Thus every such sum is at most

$$H = \sum_{d=18}^{47} \frac{1}{d} = \frac{442017301628992345493}{442720643463713815200} < 1.$$

□

For the remaining lengths, consider a state $(a/b, t, D)$, where a, b are coprime positive integers, $t \geq 1$ is the number of terms still needed, and $D \geq 1$ is the least permitted denominator. Its solutions are strictly increasing t -tuples starting at or above D with reciprocal sum a/b .

Lemma 3. *For $t \geq 2$, every first denominator d of a solution lies in*

$$\max\{D, \lfloor b/a \rfloor + 1\} \leq d \leq \lfloor tb/a \rfloor. \quad (1)$$

It must also satisfy

$$h(d, t) := \sum_{j=0}^{t-1} \frac{1}{d+j} \geq a/b. \quad (2)$$

For each retained d , the remaining problem is exactly $(a/b - 1/d, t - 1, d + 1)$.

Proof. The remaining $t - 1$ terms have positive sum, so $1/d < a/b$. Every denominator is at least d , giving $a/b \leq t/d$. Distinctness gives the sharper upper bound $h(d, t)$. Subtraction is positive by the strict first inequality. Appending d to a solution of the child state, or deleting it from a solution of the parent state, gives inverse maps. Finally, $h(d, t)$ strictly decreases with d , so its first strict failure excludes all subsequent d . Equality must be retained. $\square$

At $t = 1$, the separate test is $a = 1$ and $b \geq D$. An empty interval in (1) has no solutions. Each interval is finite; induction on t therefore proves termination and complete coverage, even though there is no fixed global cap on denominators. A straightforward exact implementation is:

```
search(r,t,D):
  write r = a/b in lowest terms
  if t == 1: return [(b,)] if a == 1 and b >= D else []
  answers = []
  for d from max(D,b//a+1) through (t*b)//a:
    if sum(1/(d+j) for j in range(t)) < r: break
    for tail in search(r-1/d,t-1,d+1):
      append (d,) + tail to answers
  return answers
```

All fractions in this pseudocode denote exact rational operations. The actual programs count solutions without retaining the entire tree and solve two-term states directly.

Lemma 4. *Solutions $D \leq x < y$ of $a/b = 1/x + 1/y$ correspond bijectively to integers q satisfying*

$$q \mid b^2, \quad \max\{1, aD - b\} \leq q < b, \quad q \equiv -b \pmod{a}. \quad (3)$$

The correspondence is

$$x = (b + q)/a, \quad y = (b + b^2/q)/a.$$

Proof. Multiplication and rearrangement give $(ax - b)(ay - b) = b^2$. Set $q = ax - b > 0$. The order $x < y$ is equivalent to $q < b$. Conversely, q satisfying (3) makes x integral and at least D . Since $\gcd(a, b) = 1$, also $\gcd(a, q) = 1$. For $a > 1$, reduction modulo a then gives $b^2/q \equiv -b \pmod{a}$, so y is integral; for $a = 1$ integrality is immediate. The factor identity proves the required reciprocal sum and strict order. $\square$

Alternatively, scan the interval (1) for x with $t = 2$, set $q = ax - b$, and require that q divide bx and $y = bx/q > x$. There is no upper truncation of y in either method. Divisor enumeration splits the prime factors of b^2 into disjoint groups and combines their divisors subject to the interval and residue condition. Every divisor has a unique such decomposition. Dropping a partial product at least b is safe, as its remaining factors are positive integers. If the attempted factorization is incomplete, the program scans the full interval instead of declaring failure.

4 Necessary prime-power coverage bounds

The 33-term tree is shortened by additional necessary conditions.

Lemma 5. *If $a/b = \sum_{i=1}^t 1/x_i$ in lowest terms and $p^e \mid b$, at least one x_i is divisible by p^e .*

Proof. The reduced denominator b divides $\text{lcm}(x_1, \dots, x_t)$. If every x_i had p -adic valuation less than e , their least common multiple would also, a contradiction. $\square$

Take any finite set Q of required prime powers dividing b . Assign each requirement to one denominator that supplies it. For $A \subseteq Q$, define

$$\ell(A) = \text{lcm}(A), \quad \ell(\emptyset) = 1, \quad m(A, D) = \ell(A) \left\lceil \frac{D}{\ell(A)} \right\rceil.$$

A denominator assigned A is at least $m(A, D)$. Put $S = 2^{24}$ and $u(A, D) = \lceil S/m(A, D) \rceil$. Define

$$\begin{aligned} B(1, Q, D) &= u(Q, D), \\ B(t, Q, D) &= \max_{A \subseteq Q} (u(A, D) + B(t-1, Q \setminus A, D)). \end{aligned}$$

Empty assigned sets are allowed.

Proposition 6. *Every admissible tail at this state satisfies $S \sum_i 1/x_i \leq B(t, Q, D)$. Consequently $aS > bB(t, Q, D)$ is a sufficient condition for exclusion.*

Proof. For an assignment supplied by Lemma 5, the scaled reciprocal of each denominator is bounded above by the corresponding u . The recursion maximizes the sum over all assignments. Dropping distinctness only increases this bound, and every integer rounding is upward. $\square$

The implemented groups of possible requirements are

$$\begin{aligned} &(16, 32, 64, 9, 27, 81, 25, 49), \\ &(11, 13, 17, 19, 23, 29, 31, 37), \\ &(41, 43, 47, 53, 59, 61, 67, 71). \end{aligned}$$

Only entries dividing the current reduced denominator are required. Each group gives a valid bound, so their minimum is valid. Overlapping powers use least common multiples, not products. Omitted prime powers weaken pruning and impose no restriction on solutions. The additional dynamic-programming test is used for $t = 4, 5, 6$ and $D < 1890$; outside its supported domain the original recurrence remains available.

For three-term states, after choosing d , remove requirements already dividing d and assign the others to two denominators at least $d+1$. For a partition $(A, Q \setminus A)$, put

$$q_1 = \ell(A), \quad q_2 = \ell(Q \setminus A), \quad x = m(A, d+1), \quad y = m(Q \setminus A, d+1).$$

If $x \neq y$, the assignment's maximum reciprocal sum is $1/x + 1/y$. If $x = y = z$, distinctness forces one denominator to its next allowed multiple, giving maximum

$$\frac{1}{z} + \frac{1}{z + \min(q_1, q_2)}.$$

Both alternatives are achieved under the relaxed assignment constraints. Maximize over all partitions and round S times this maximum upward to an integer P . The necessary test is

$$(ad - b)S \leq bdP.$$

Its failure skips only this particular d . This modular bound is not assumed monotone; only the harmonic test ends the loop.

5 Arithmetic safety and exact computation

The implementation uses GMP integers and rationals [2], with guarded fixed-width shortcuts. No floating-point quantity decides a branch deletion. The group least common multiples are respectively

$$6350400, \quad 35336848261, \quad 75186702453419,$$

all below 2^{64} . Every subset least common multiple divides its group value L . For precomputation at $D \leq 1890$, a minimum multiple is at most $L + D$, and the distinctness correction adds at most L . Thus the product of the two corrected denominators is bounded by $(L + 1890)(2L + 1890)$. This bound, and the ceiling-division numerator $(x + y)S + xy - 1$, are below 2^{128} for each group. This universal estimate also covers overlapping prime powers.

Each $u(A, D)$ is at most S . At most six such bounds are added in a cached entry, so unsigned 32-bit totals suffice. Cache indices are checked. The native three-term path is entered only after GMP verifies that

$$ah(h + 1)(h + 2), \quad b(3h^2 + 6h + 2)$$

have at most 127 bits, where h is the upper endpoint of the first-denominator interval. These bound the positive polynomial intermediates for $d \leq h$, including the harmonic comparison and the products needed for subtraction. The lower endpoint ensures $ad - b > 0$. Two-term multiplication guards bound ax and bx throughout the scan interval. Unsafe operations use GMP instead. Coverage comparisons have explicit overflow checks; an unsupported comparison retains the branch. Rational subtraction is reduced exactly, and native remainders are reduced by a greatest common divisor before conversion.

A useful bound on factorization inputs follows without restricting the last two denominators. In any 33-term solution, the first 30 reciprocals sum to at most H from Lemma 2. Therefore

$$1 - H \leq \frac{1}{n_{31}} + \frac{1}{n_{32}} + \frac{1}{n_{33}} \leq \frac{3}{n_{31}}, \quad n_{31} \leq \left\lfloor \frac{3}{1 - H} \right\rfloor = 1888.$$

Every denominator in the 31-term prefix is thus at most 1888, and its reduced remainder denominator has no larger prime factor. Neither n_{32} nor n_{33} is bounded by 1888.

5.1 Completed exclusions

Each root is $(1, t, 18)$, without a prescribed prefix. A separately written rational solver, without coverage pruning or the native shortcuts, completed the following searches:

t	Nodes	Two-term first-denominator scans	Solutions
31	20,293	69,727	0
32	14,678,224	114,603,091	0

The reviewed optimized program was also rebuilt and replayed for all three lengths. The full 33-term replay completed normally with the following totals:

Quantity	Count
Visited nodes	201,158,430
Two-term states	108,965,614
Scanned first denominators	13,238,452,833
Factor closures	3,275
Solutions	0

Its exit code was zero and its standard error was empty. These figures agree with the original completed traversal. A separate producer using the same coverage recurrence serialized the wide two-term obligations. All 3,275 distinct obligations were checked by newly written Python code: it reconstructed each exact prefix, reduced remainder and lower denominator bound, trial-factored the remainder denominator using independently tested primes, and solved (3). It did not consume the producer’s saved closure factors or conclusions. Every obligation had no completion.

Independent bound checks compared four outputs on 4,608 combinations of lower bound, group and subset: all 18,432 exact comparisons agreed. Additional exact searches compared solution counts on 154 states, covering one through six terms, impossible harmonic tails, random small inputs, cache boundaries, and positive examples with final denominator 10^{100} . These are implementation controls, not an exhaustive correctness proof by testing. Their role supplements the coverage and arithmetic arguments. Completion receipts, rather than elapsed times or partial progress, establish that the specified computations terminated.

Proof of Theorem 1. The integer identity proves $S_{34} \neq \emptyset$. Lemma 2 excludes all lengths at most 30. Lemmas 3 and 4 prove that the completed rational traversals cover lengths 31 and 32. Their solution counts are zero. Lemma 5, Proposition 6, and the distinct two-tail bound justify every additional deletion used for length 33. The guarded arithmetic implements these necessary inequalities exactly. The completed full-root traversal, with the terminal computations checked as described above, returns no solution. Thus S_t is empty for every positive $t < 34$, proving $k(18) = 34$. $\square$

6 Reproducibility and verification boundary

The accompanying source package retains the exact witness, reviewed traversal, separately written lower-bound solver, serialized terminal inputs, independent checker, test harnesses, and complete outputs. Its reproduction guide identifies their locations and commands. The audited environment used Apple clang 16.0.0 with C++17 and optimization level O3, GMP 6.3.0, and Python 3.9.6. C++ assertions were enabled by the recorded build commands. Reproduction requires compiling the search executable and invoking it separately with arguments 31 18, 32 18, and 33 18, with no fixed prefix. The lower solver and independent terminal checker are separate invocations. The guide also distinguishes replaying the full tree from merely checking stored results.

Comparable exact-search practice in other finite number-theoretic problems appears in a certified extension of a divisor-function search frontier [4] and in a complete pairwise-coprime grid threshold computation with two independent enumerators [3]. Those papers address different extremal questions and supply no premise for the Egyptian-fraction exclusion proved here.

Discovery included exploratory searches, local replacement searches for shorter identities, and trials of alternative arithmetic optimizations. Those experiments are not proof premises. Exact certification consists of the integer identity, proved coverage inequalities, completed searches, and terminal recalculation. The full 33-term replay shares the reviewed traversal source. Independence here refers to the separate lower solver, terminal checker, bound and state controls, and a fresh adversarial reconstruction of the argument; it does not mean that two unrelated complete algorithms excluded length 33.

The proof trusts the reviewed programs, compiler, GMP, Python integer arithmetic, and ordinary execution hardware. It is not a proof-assistant formalization or a compact formal certificate for the entire search tree. The smaller controls do not remove that trust boundary. No timeout, incomplete adaptive search, or unsupported numerical pattern contributes to the exclusion.

7 Limitations and assistance disclosure

The result determines one finite minimum. It supplies neither a classification of all 34-term expansions nor a uniform bound resolving the asymptotic problem. The first denominator remains free to exceed 18 throughout the exclusion. Historical statements are attributed to the dated online report; the original book page was inaccessible to the project. The literature search was limited and establishes no priority claim. The independent computational review was a project verification step, not journal peer review or journal acceptance.

OpenAI Codex language-model agents assisted with mathematical exploration, program generation and revision, verification work, and manuscript preparation. A separate language-model context performed the recorded adversarial audit and computational checks. That separation does not constitute an independent human referee assessment. Model-generated arguments and confidence are not evidence for the theorem: its support is the explicit mathematics and reproducible exact computations described here. The retained records do not establish a separate human line-by-line verification, and no such verification is claimed.

References

- [1] Erdős #295: $k(17) = 32$, and now $33 \leq k(18) \leq 35$ by exact exhaustion. Erdős Problem a Day, <https://erdosproblemaday.com/day/295-k17-exact>, August 2026. Dated 11 August 2026; accessed 5 September 2026. Online computational report; site credits Patrick White and Claude (Anthropic).
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- [3] Zijian Zeng. The exact two-row pairwise-coprime grid threshold is $L_{41} = 529$. SSRN preprint, 2026.
- [4] Zijian Zeng. A strict certified search extension for Erdős problem 647. SSRN preprint, 2026.